

Tutorial set 3

1. For a quantum harmonic oscillator, show that

$$\langle n' | X | n \rangle = \sqrt{\left(\frac{\hbar}{2m\omega}\right)} [\delta_{n',n+1} \sqrt{n+1} + \delta_{n',n-1} \sqrt{n}]$$

$$\langle n' | P | n \rangle = \sqrt{\left(\frac{m\omega\hbar}{2}\right)} i [\delta_{n',n+1} \sqrt{n+1} - \delta_{n',n-1} \sqrt{n}]$$

Calculate this using two methods. (a) Use the wave functions in the x basis. (b) Use the $|n\rangle$ basis.

2. Find $\langle X \rangle$, $\langle P \rangle$, $\langle X^2 \rangle$, $\langle P^2 \rangle$, $\Delta X \cdot \Delta P$ in the state $|n\rangle$.

3. Show that

$$\langle n | X^4 | n \rangle = \left(\frac{\hbar}{2m\omega}\right)^2 [3 + 6n(n+1)]$$

4. At $t = 0$ a particle starts out in $|\psi(0)\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. Find $\psi(t)$. Find $\langle X(0) \rangle$, $\langle P(0) \rangle$, $\langle X(t) \rangle$ and $\langle P(t) \rangle$.

5. Find the expectation value of the potential energy in the n th state of the harmonic oscillator.

6. In the ground state of the harmonic oscillator, what is the probability of finding the particle outside the classically allowed region ?